The augmentation module of the Steiner system on 23 points

Working manuscript: independent review pending

This note completes the length-253 instance of the common incidence argument. Its source is Leemans and Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Math. Commun. 21 (2016), 239–249 (L050), Theorem 1(d) and Proposition 5. We retain the source's Steiner construction and the exact weight enumerations already completed for this construction. The remaining irreducibility and full automorphism assertions follow from the same two certificate arguments used for the length-120 and length-176 cases. The finite programs and their literal inputs are necessary proof premises. No historical novelty or full Lean formalization is asserted.

Let $\mathcal{D}$ be the specified $S(4,7,23)$, with point set Ω and 253 heptads. Its point–block incidence matrix is H. Two distinct heptads intersect in one or three points, and Γ joins them in the latter case. Write A for its adjacency matrix. All vector spaces in this note are over $F = \mathbb{F}_3$. Put

$$E = \{x \in F^{23} : \textstyle\sum_i x_i = 0\}, \quad C = \operatorname{im} A, \quad D = H^{\mathsf{T}}E.$$

The shared construction checks the Steiner condition and actual incidence matrix, not merely the strongly regular parameters.

Theorem 1. *The code C is the orthogonal direct sum $\langle \mathbf{j} \rangle \oplus D$, where D is an absolutely irreducible 22-dimensional module for M_{23}. Both C and D are LCD. Their parameters and minimum multiplicities are*

code	*parameters*	*minimum words*
C	$[253, 23, 77]_3$	46
D	$[253, 22, 97]_3$	3542
$C^{\perp}$	$[253, 230, 6]_3$	850080
$D^{\perp}$	$[253, 231, 6]_3$	850080.

The full groups of coordinate permutations preserving C and Γ are equal and isomorphic to M_{23}.

Proof. There are 77 heptads through each point and 21 through each pair. Consequently

$$HH^{\mathsf{T}} = 56I_{23} + 21J_{23} = 2I_{23} \quad \text{over } F.$$

In particular H^{T} is injective. Every column sum is seven, so $H^{\mathsf{T}}\mathbf{1} = \mathbf{j}$, and the two intersection sizes give

$$A = J_{253} - H^{\mathsf{T}}H = H^{\mathsf{T}}(J_{23} - I_{23})H.$$

The middle matrix is invertible: its eigenvalues on the constant line and its sum-zero complement are $22 = 1$ and -1 in F. Thus $C = \operatorname{im} H^{\mathsf{T}}$, of dimension 23. Since $23 \neq 0$ in F, the point space decomposes orthogonally into its constant line and E. The transferred form is twice the ordinary dot product. It is nondegenerate on both summands. Their images give the asserted orthogonal splitting and both LCD conclusions.

We next prove the irreducibility, using explicit matrices in place of a modular character-table inference. The five permutations of Ω in `generators.json` preserve the full heptad set. They

therefore act on the code coordinates and on E. In the basis $e_i - e_{23}$, $1 \le i \le 22$, the row matrix for a point permutation g has entries

$$M(g)_{ij} = \mathbf{1}_{j=g(i)} - \mathbf{1}_{j=g(23)}, \qquad 1 \le i, j \le 22.$$

An indicator with point 23 is zero because $j \le 22$. The certificate checks that multiplication by these matrices agrees with the induced permutation of all 253 coordinates on the actual rows $h_i - h_{23}$. It produces 484 independent words in these matrices, each of length at most six, and reevaluates every word before reducing the flattened matrices. They span $M_{22}(F)$.

Any invariant subspace of E is therefore stable under every endomorphism of E. A nonzero vector can be sent to any prescribed vector by an endomorphism, so such a subspace must be all of E. The same independent matrix words remain independent after any field extension. This proves absolute irreducibility of E, and hence of its equivariant image D. The argument is the general full-matrix-algebra criterion already used for the other residual codes.

The existing enumeration records contain the complete weights of the row-reduced adjacency code and of the difference code. Their bases are compared byte for byte with the bases reconstructed here. The latter row space is D: its dimension is 22, it lies in C and is orthogonal to $\mathbf{j}$. The recorded enumerations have respectively 3^{23} and 3^{22} coefficient vectors. They give the two distances and counts in the first two rows of the table. The dual distributions follow from the classical ternary MacWilliams identity

$$A_j^{\perp} = 3^{-k} \sum_{i=0}^{253} A_i \sum_{s} (-1)^s 2^{j-s} \binom{i}{s} \binom{253-i}{j-s}.$$

All coefficients are evaluated over the integers. Nonnegativity, integrality and total 3^{253-k} are checked. Both duals have no words of weights one through five and exactly 850080 of weight six. This proves the table without repeating either exhaustive enumeration.

It remains to prove that the displayed automorphism subgroup is full. Each point-incidence row h_i is a word of C of weight 77. The 46 words $h_i, -h_i$ are distinct. By the complete minimum count they are all minimum words. Their 23 supports are exactly the sets of heptads through the respective points. A code automorphism permutes these supports and thus induces an incidence-preserving permutation of the points and blocks of $\mathcal{D}$. Conversely every design automorphism permutes the incidence rows and preserves C. Hence $\mathrm{Aut}(C)$ is the group induced on the blocks by $\mathrm{Aut}(\mathcal{D})$.

For an explicit upper bound, form the bipartite point–block incidence graph, with its two parts initially given different colours. Refine the colours by the old colour and the counts of neighbours of each colour. After individualizing points $1, 2, 3, 4, 5, 6$ successively, the certificate obtains current cell sizes

$$23, \quad 22, \quad 21, \quad 20, \quad 16, \quad 3.$$

All points are then distinguished. A pointwise automorphism fixes every heptad, since distinct heptads have different point neighbourhoods. Each cell contains the orbit of its chosen point under the preceding stabilizer; repeated orbit–stabilizer therefore gives

$$|\mathrm{Aut}(\mathcal{D})| \le 23 \cdot 22 \cdot 21 \cdot 20 \cdot 16 \cdot 3 = 10200960.$$

Exact group computations on the literal generators give a subgroup of this order and an isomorphism to the standard group M_{23}. The induced block action is checked to be faithful. Thus the upper bound is attained and identifies the full design and code groups. Finally the subgroup preserves Γ, and every graph automorphism preserves its adjacency row space. Therefore

$G \leq \mathrm{Aut}(\Gamma) \leq \mathrm{Aut}(C) = G$, proving the asserted equality. The upper bound uses only incidence refinement; graph-isomorphism software was used to discover the generators and is not a final premise. □

Qualifications to Proposition 5. The main Theorem 1(d) is recovered as stated. Proposition 5 additionally assigns distance eight to $D^\perp$ and declares it irreducible. Both assertions are false for these codes. Since $D \subset C$, the invariant space $C^\perp$ of dimension 230 is a proper nonzero subspace of $D^\perp$, whose dimension is 231. The same inclusion already gives $d(D^\perp) \leq d(C^\perp) = 6$; the exact value six is verified above. The proposition also calls D the "unique and smallest" irreducible module without qualifications. The unqualified statement overlooks the trivial one-dimensional module. Its intended classification requires an input separate from our matrix certificate. We retain the complete 3-modular character table of M_{23}, as in the source's reference [13], Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). Its irreducible degrees over an algebraic closure are

$$1, 22, 45, 45, 104, 104, 231, 253, 770, 770, 990, 990, 1035.$$

The accompanying GAP calculation checks these entries against its installed table `M23mod3`; it does not reprove completeness of that table. Together with standard scalar extension and descent for simple modules over finite fields, the table gives a unique 22-dimensional irreducible F-module, and 22 is the least nontrivial degree. Indeed a simple module splits after scalar extension into an orbit of absolutely simple constituents of equal degree. Below degree 22 the only available constituent is trivial, which forces the original simple module to be trivial. At degree 22 the unique constituent is the one already realized by D; descent of module isomorphisms gives uniqueness over F. These are retained classification premises, explicitly distinct from the direct proof of irreducibility of D. The source's open uniqueness question for the strongly regular parameters $(253, 140, 87, 65)$ is not resolved here.